Technical Document #127: Blockchain - Comprehensive Overview

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Non-fungible tokens (NFTs) represent unique digital assets on the blockchain. They have transformed digital ownership and art markets.

Smart contracts are self-executing contracts with terms directly written into code. They automatically enforce agreements without intermediaries.

Proof of Stake (PoS) selects validators based on their stake in the network. It is more energy-efficient than Proof of Work.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Key Takeaways:

- Zero-knowledge proofs allow one party to prove knowledge of information without revealing the information itself.
- Blockchain is a distributed ledger technology that records transactions across many computers.
- Decentralized finance (DeFi) recreates traditional financial systems using blockchain technology.